Pengcheng Wang

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Overview

- Highly analytical and detail-oriented software engineer with a strong foundation in software development and mathematics.
- Proactive team player with experience managing multiple projects and consistently meeting tight deadlines.
- Quick to adapt and technically proficient across a wide range of modern programming languages and databases.

Education

University of Toronto *Sept 2020 – May 2024*
Bachelor of Computer Science in Computer Science

- GPA: 3.72/4.0, Dean's List (2022 - 2024)
- **Coursework:** Data Structures, Algorithms, Operating Systems, Object-Oriented Programming, Database, Software development, Networking Security

University of Waterloo *Sept 2024 – Sept 2025*
Master of Engineering in Elec&Comp Eng.

- GPA: 3.66/4.0
- **Coursework:** Algorithms, Networking Security, Software development, Machine Learning

Skills

Languages: Python, Java/Clojure, JavaScript/Typescript, Swift, C, SQL

Databases: SQLite, PostgreSQL, MongoDB, Neo4J

Projects

Quick Agent | *Clojure, Python, Swift, JS, Ollama/MLX* *GitHub Repo*

- Built a modular personal assistant system using Clojure as the core logic layer and Babashka Pods for cross-language interoperability, inspired by MCP.
- Integrated local LLMs (via Ollama and MLX) to process user intents and generate responses in real time.
- Enabled natural user interaction through iMessage, routed into multi-agent workflows (e.g. web search, event scheduling and etc.).
  - Developed lightweight Swift-based Pods to access macOS APIs like Calendar (EventKit) and iMessage (AppleScript).
  - Leveraged Python and JavaScript Pods for advanced NLP and system integration using high-performance libraries.

FlooringTerminal | *Svelte, Typescript, TailwindCSS* *Company Website*

- Designed and developed a modern company website for a flooring retailer using Svelte, TypeScript, and Tailwind CSS.
- Implemented responsive UI components, dynamic product listings, and contact forms to enhance customer engagement.
- Deployed the site on Cloudflare Pages for optimal performance and reliability.

Auto Ambient Brightness Recognition | *Python, Pytorch, PyQt* *GitHub Repo*

- Developed a software solution for Windows laptops without built-in light sensors to automatically adjust screen brightness based on ambient light.
- Used a Convolutional Neural Network (CNN) with OpenCV to analyze ambient lighting via webcam, adjusting display settings in real time.
- Built a user-friendly GUI in PyQt and packaged the application for end-users.

TheRide | *Typescript (MongoDB, Express.js, React.js, Node.js), Socket.io* *GitHub Repo*

- Designed and built an online Q&A platform enabling college students to seek and provide academic help in real time.
- Implemented the full-stack web application using MERN stack (MongoDB, Express.js, React.js, Node.js) and Socket.io for instant messaging.
- Delivered core features including user-generated questions, answers, and real-time communication; received top grade for project delivery.

Zoomer Rideshare | *Java, PostgresSQL, MongoDB, Neo4j* *GitHub Repo*

- Engineered a ridesharing platform that matches users with nearby drivers for safe and efficient transportation.
- Architected with microservices, PostgreSQL, MongoDB, and Neo4j to manage users, trips, and route optimization.
- Integrated all components using Docker and exposed services via an API gateway for frontend access.